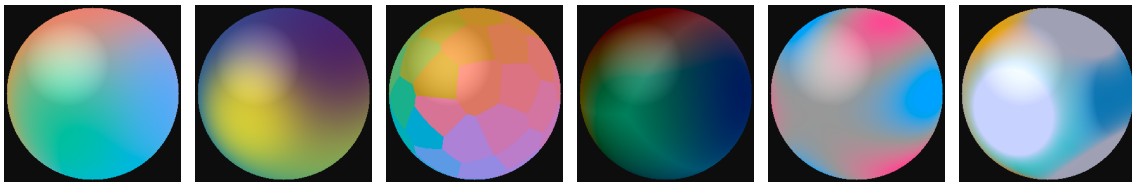


Principia

Spherical Colour Map System

Implementation Specification



This document specifies the mathematical foundations and implementation details for the two-artefact spherical colour map explorer developed for the Principia three-body visualiser project. It covers colour science, the von Mises–Fisher blending engine, all colour map and pattern modes, the colour-vision deficiency simulation pipeline, and the shape sphere physics overlay.

Version 1.0 • Artefacts 01_colour_maps and 02_patterns_special

Written in conjunction with the Principia three-body fractal renderer.

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Foundations

The Encoding Problem

The goal is to assign a colour to every point on the unit sphere $S^2 \subset \mathbb{R}^3$. Each point is a unit normal vector $\hat{\mathbf{n}} = (n_x, n_y, n_z)^T$ with $\|\hat{\mathbf{n}}\| = 1$. We require:

1. **Continuity.** Nearby points on S^2 receive similar colours. No discontinuities or seams.
2. **Perceptual uniformity.** Equal angular distances on S^2 correspond to equal perceived colour differences.
3. **Reserved lightness dimension.** Brightness (L) is reserved for an independent scalar field (e.g. orbital stability). The colour map must operate at fixed L .
4. **Full 2-D encoding.** Both degrees of freedom (latitude and longitude) must be discriminable from colour alone.

Colour Spaces

Linear sRGB

The display pipeline works in linear sRGB. The gamma expansion and compression functions are:

$$\gamma(v) = \begin{cases} v/12.92 & v \leq 0.04045 \\ \left(\frac{v+0.055}{1.055}\right)^{2.4} & v > 0.04045 \end{cases} \quad (1)$$

$$\Gamma(c) = \begin{cases} 12.92 c & c \leq 0.0031308 \\ 1.055 c^{1/2.4} - 0.055 & c > 0.0031308 \end{cases} \quad (2)$$

OKLAB

OKLAB (Ottosson, 2020) is a perceptually uniform colour space with axes $L \in [0, 1]$ (lightness), a (green–red), and b (blue–yellow). The conversion from OKLAB to linear sRGB is:

$$\begin{pmatrix} \ell' \\ m' \\ s' \end{pmatrix} = M_1 \begin{pmatrix} L \\ a \\ b \end{pmatrix}, \quad \begin{pmatrix} \ell \\ m \\ s \end{pmatrix} = \begin{pmatrix} \ell'^3 \\ m'^3 \\ s'^3 \end{pmatrix}, \quad \begin{pmatrix} R_\ell \\ G_\ell \\ B_\ell \end{pmatrix} = M_2 \begin{pmatrix} \ell \\ m \\ s \end{pmatrix} \quad (3)$$

where

$$M_1 = \begin{pmatrix} 1 & 0.3963 & 0.2158 \\ 1 & -0.1056 & -0.0639 \\ 1 & -0.0895 & -1.2915 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 4.0767 & -3.3077 & 0.2310 \\ -1.2684 & 2.6098 & -0.3413 \\ -0.0042 & -0.7034 & 1.7076 \end{pmatrix}$$

In polar form: $a = C \cos \theta_h$, $b = C \sin \theta_h$, where $C \geq 0$ is the *chroma* and θ_h is the *hue angle*. This (OKLCH) parameterisation is used throughout.

The von Mises–Fisher Blending Engine

Von Mises–Fisher Distribution

The von Mises–Fisher (vMF) distribution on S^2 is the spherical analogue of a Gaussian:

$$f(\hat{\mathbf{n}}; \hat{\mathbf{p}}, \kappa) \propto \exp(\kappa \hat{\mathbf{n}} \cdot \hat{\mathbf{p}}) \quad (4)$$

Maximum at $\hat{\mathbf{n}} = \hat{\mathbf{p}}$, decaying with geodesic distance. The concentration κ controls spread.

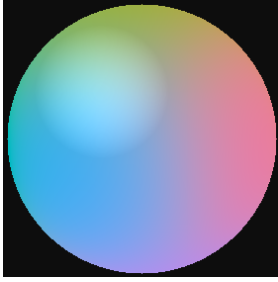
6-Pole OKLAB Colour Map

Place $N = 6$ colour poles at the axis intersections $\{\pm\hat{x}, \pm\hat{y}, \pm\hat{z}\}$ with opposing poles assigned complementary hues. The (a, b) coordinates at query point $\hat{\mathbf{n}}$ are a vMF-weighted mean in OKLAB:

$$a(\hat{\mathbf{n}}) = C \cdot \frac{\sum_{i=1}^6 w_i \cos \theta_i}{\sum_{i=1}^6 w_i}, \quad b(\hat{\mathbf{n}}) = C \cdot \frac{\sum_{i=1}^6 w_i \sin \theta_i}{\sum_{i=1}^6 w_i}, \quad w_i = \exp(\kappa \hat{\mathbf{n}} \cdot \hat{\mathbf{p}}_i) \quad (5)$$

Two hue schemes are implemented: the full OKLAB scheme and the Okabe–Ito CB-safe scheme:

Pole	Direction	Full OKLAB Colour	θ_i	Okabe–Ito θ_i
$\hat{\mathbf{p}}_1$	$+\hat{x}$	Red	0°	250° (Blue)
$\hat{\mathbf{p}}_2$	$-\hat{x}$	Cyan	180°	70° (Orange)
$\hat{\mathbf{p}}_3$	$+\hat{y}$	Green	120°	30° (Vermillion)
$\hat{\mathbf{p}}_4$	$-\hat{y}$	Magenta	300°	210° (Sky Blue)
$\hat{\mathbf{p}}_5$	$+\hat{z}$	Blue	240°	170° (Bluish Green)
$\hat{\mathbf{p}}_6$	$-\hat{z}$	Yellow	60°	350° (Reddish Purple)



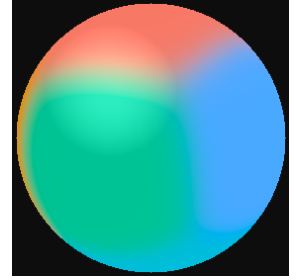
(a) VMF OKLAB ($\kappa = 3$)



(b) VMF Okabe–Ito ($\kappa = 3$)



(c) Okabe–Ito ($\kappa = 1$)



(d) Okabe–Ito ($\kappa = 10$)

Figure 1: VMF colour maps. Left to right: full OKLAB hue scheme; Okabe–Ito hue scheme; effect of low κ (wide blobs); effect of high κ (crisp poles). Note how opposing poles cancel to neutral grey at midpoints.

Key properties of Eq. (5): seamless (no `atan2`); opposing poles cancel at their midpoint; linear in OKLAB ensuring perceptual uniformity; fully parameterised by L , C , κ .

Seamless LUT Sphere

To apply a 1-D colour LUT seamlessly to S^2 , we use the same vMF engine. $N_e = 16$ colours are sampled from the LUT at uniform intervals and placed as equatorial poles at $\hat{\mathbf{p}}_i^{\text{eq}} = (\cos(2\pi i/N_e), \sin(2\pi i/N_e), 0)$; the LUT endpoints are placed at the north and south poles. The blend is then evaluated identically to Eq. (5) but in RGB space.

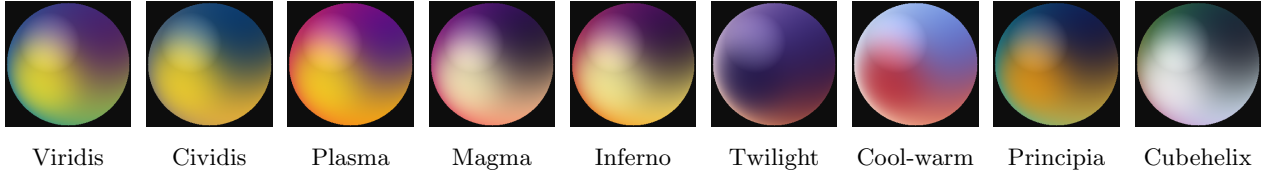


Figure 2: All LUT-based colour maps rendered via seamless equatorial VMF poles. No seam is visible on any of these spheres.

The **Twilight** LUT is cyclic (start = end colour), making it ideal for longitude encoding. The equatorial-poles approach exploits this automatically. **Cubehelix** is generated analytically: hue spirals with monotone-increasing lightness, making it CB-tolerant by construction.

Colour-Vision Deficiency Simulation

CVD simulation is applied as a post-process in *linear* sRGB, after all pixel computation. Matrix M_{cvd} multiplies the linear (R_ℓ, G_ℓ, B_ℓ) triplet:

$$\begin{aligned}
 M_{\text{deutan}} &= \begin{pmatrix} 0.625 & 0.375 & 0 \\ 0.700 & 0.300 & 0 \\ 0 & 0.300 & 0.700 \end{pmatrix} & M_{\text{protan}} &= \begin{pmatrix} 0.567 & 0.433 & 0 \\ 0.558 & 0.442 & 0 \\ 0 & 0.242 & 0.758 \end{pmatrix} \\
 M_{\text{tritan}} &= \begin{pmatrix} 0.950 & 0.050 & 0 \\ 0 & 0.433 & 0.567 \\ 0 & 0.475 & 0.525 \end{pmatrix} & M_{\text{achrom}} &= \begin{pmatrix} 0.299 & 0.587 & 0.114 \\ 0.299 & 0.587 & 0.114 \\ 0.299 & 0.587 & 0.114 \end{pmatrix}
 \end{aligned}$$

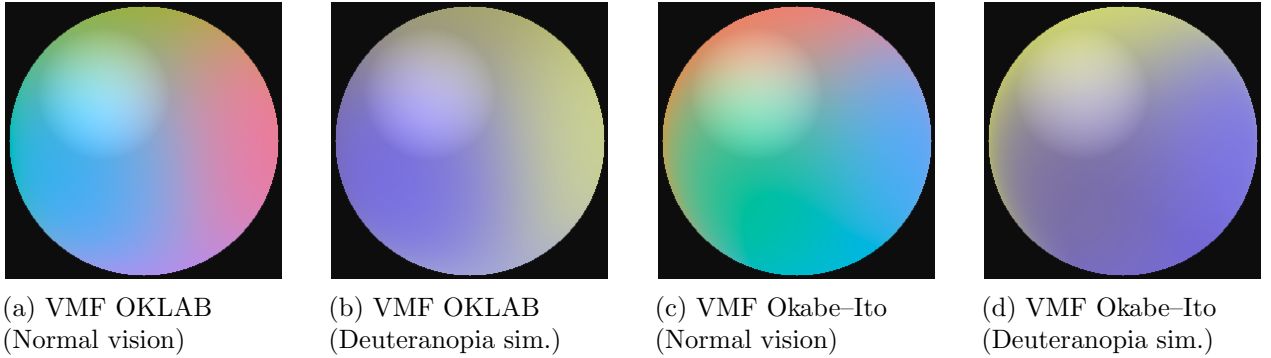


Figure 3: CVD simulation comparison. Under deuteranopia the Full OKLAB map loses the red–green distinction, collapsing two poles to near-identical orange-brown. The Okabe–Ito scheme retains all 6 pole colours because it avoids the red–green axis entirely.

Pipeline order: **pixel function** → **physics overlay** → **CVD simulation** → **canvas write**.

Rendering Pipeline

Sphere Projection

For canvas width W , height H , sphere radius $R = W/2 - 4$, and centre $(c_x, c_y) = (W/2, H/2)$:

$$s_x = (p_x - c_x)/R, \quad s_y = -(p_y - c_y)/R, \quad s_z = \sqrt{\max(0, 1 - s_x^2 - s_y^2)} \quad (6)$$

Pixels with $s_x^2 + s_y^2 > R^2$ are left transparent.

Rotation and Equirectangular Projection

The rotation matrix $M_{\text{rot}} = R_x(\beta) R_y(\alpha)$ maps screen-space normals to world space. The equirectangular projection maps $(\phi, n_z) \in [-\pi, \pi] \times [-1, 1]$ to canvas pixels, independent of rotation, and is used to expose seams and discontinuities.

Colour Map Modes (Artefact 1)



Figure 4: Additional colour map modes. Direction cosines maps each Cartesian component to a separate RGB channel; note the non-uniform lightness. Cool-warm is a diverging map ideal for bipolar fields. The Principia palette uses a custom indigo→teal→gold sequence.

Global controls: **Invert** ($v \mapsto 255 - v$), **Blend** (linear mix of any two modes), **Auto-rotate**.

Pattern Modes (Artefact 2)

All pattern modes accept a palette argument and have signature $(\hat{n}, \text{params}, \text{palette}) \mapsto [R, G, B]$.

Voronoi-Type Patterns

Octant

Partition by sign: $\text{index} = 4[n_x \geq 0] + 2[n_y \geq 0] + [n_z \geq 0]$.

Voronoi 6

Nearest axis pole: $i^* = \arg \max_i \hat{\mathbf{n}} \cdot \hat{\mathbf{p}}_i$.

Hemispheres

Dominant axis determines colour, with two shades per axis for sign.

Icosahedral

12 Voronoi cells from the icosahedron vertices $V = \{(0, \pm 1, \pm \varphi), (\pm 1, \pm \varphi, 0), (\pm \varphi, 0, \pm 1)\} / \|(0, 1, \varphi)\|$, where $\varphi = (1 + \sqrt{5})/2$.

Soft Voronoi

Sigmoid blend between two nearest poles: $t = \sigma(k_s(d_1 - d_2))$.

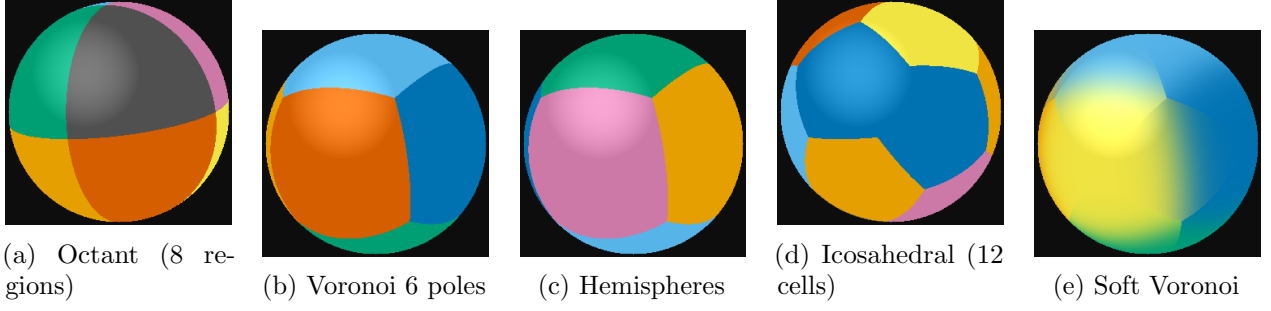


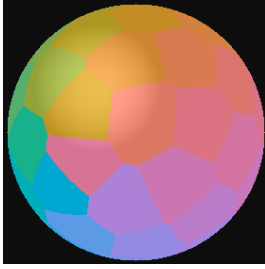
Figure 5: Voronoi-type patterns. From hard octant partitioning to sigmoid-blended soft Voronoi. Palette: Okabe–Ito (CB-safe).

Fibonacci Lattice and Dot Lattice

The golden-angle Fibonacci lattice generates N near-uniformly distributed points. For $i = 0, \dots, N - 1$:

$$n_{z,i} = 1 - 2i/(N - 1), \quad r_i = \sqrt{1 - n_{z,i}^2}, \quad \phi_i = \pi(\sqrt{5} - 1) \cdot i$$

The palette is generated with the golden-angle hue spacing $\Delta\theta = 2\pi/(1/\phi_g)$ where $\phi_g = (\sqrt{5} - 1)/2$, ensuring adjacent cells always have highly contrasting hues. The **dot lattice** variant renders points as coloured dots (angular radius $\rho = \cos(1.4/\sqrt{N})$) on a dark background.



(a) Fibonacci lattice ($N = 48$)



(b) Dot lattice ($N = 48$)

Figure 6: Fibonacci-based modes. The lattice covers the sphere with near-uniform cell areas and no polar clustering. Golden-angle palette ensures adjacent cells are always strongly distinct in hue.

Stripe Patterns

Checkerboard

Using spherical coordinates $\theta = \arccos(n_z)$, $\phi = \text{atan2}(n_y, n_x) + \pi$:

$$\text{even} = \left(\lfloor f\theta/\pi \rfloor + \lfloor f\phi/(2\pi) \rfloor \right) \bmod 2$$

Seam-free for integer f (tiles close exactly at the antimeridian).

Latitude / Longitude Stripes

$$\text{lat} : \cos(f \arccos n_z) > 0 \quad \text{lon} : \sin(f \text{atan2}(n_y, n_x)) > 0$$

Latitude stripes never use **atan2**. Longitude stripes are seamless because $\sin(n\pi) = 0$ for integer n .

Truchet Mosaic

Each patch cell (c_i, c_j) receives a deterministic diagonal split via hash $h(c_i, c_j) = \sin(127.1c_i + 311.7c_j) \cdot 43758.5 \bmod 1$. The colour is assigned based on which side of the diagonal the local coordinate (u, v) falls.

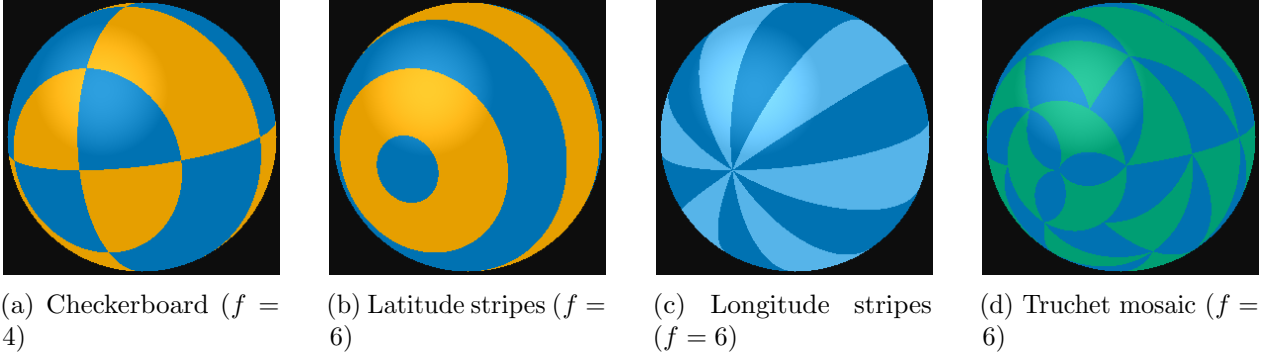


Figure 7: Stripe and mosaic patterns. All are seamless by construction. The Truchet mosaic produces interlocking maze-like patterns from a simple deterministic hash.

Overlay Modes

Grid Overlay

Replace pixels within $\epsilon = 0.04$ of any grid line with a user-chosen line colour. The underlying map is unaffected elsewhere.

Iso-Hue Contours

Contour lines trace directions of constant VMF hue angle $\psi = \text{atan2}(b_{\text{vmf}}, a_{\text{vmf}})$, spaced uniformly in hue.

Gradient Magnitude

Finite-difference approximation of colour change rate:

$$|\nabla \mathbf{c}| \approx \frac{1}{2} \sqrt{\|\mathbf{c}(\hat{\mathbf{n}} + \varepsilon \hat{x}) - \mathbf{c}(\hat{\mathbf{n}})\|^2 + \|\mathbf{c}(\hat{\mathbf{n}} + \varepsilon \hat{y}) - \mathbf{c}(\hat{\mathbf{n}})\|^2}$$

Bright regions correspond to basin boundaries. On a fractal attractor the fractal boundary structure would emerge directly.

Perlin Noise

4-octave 3-D value noise in Cartesian coordinates, ensuring no seam:

$$v(\hat{\mathbf{n}}) = \sum_{k=0}^3 \frac{1}{2^k} \omega(2^k s n_x, 2^k s n_y, 2^k s n_z) \Big/ \sum_{k=0}^3 \frac{1}{2^k}$$

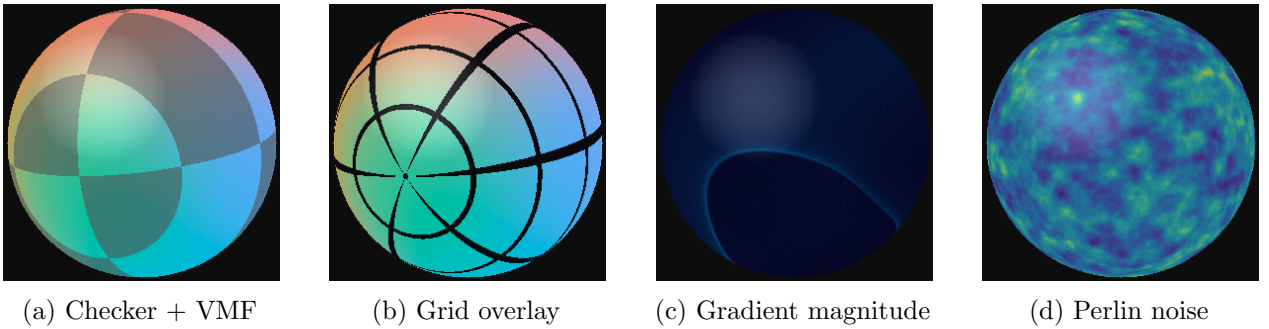


Figure 8: Overlay modes. Checker+VMF combines discrete structure with continuous position encoding. Gradient magnitude highlights the basin boundary geometry — on a fractal this would reveal the fractal dimension directly.

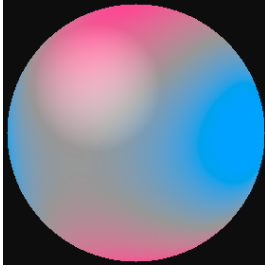
Special Modes

Real Spherical Harmonics

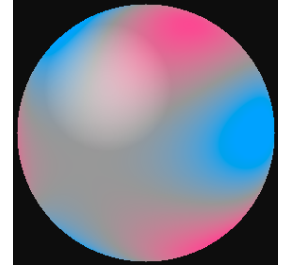
The real spherical harmonics $Y_\ell^m : S^2 \rightarrow \mathbb{R}$ encode the multipole structure of the sphere. The normalised value $v = Y_\ell^m / \max |Y_\ell^m|$ is mapped to colour by blending between user-chosen positive and negative lobe colours proportionally to $|v|$, with neutral grey at $v = 0$ (nodal lines).

Selected forms for $\ell \in \{1, 2, 3\}$:

$$\begin{aligned} Y_1^0 &= \sqrt{3/4\pi} n_z, & Y_1^1 &= \sqrt{3/4\pi} n_x \\ Y_2^0 &= \sqrt{5/16\pi} (2n_z^2 - n_x^2 - n_y^2), & Y_2^2 &= \sqrt{15/16\pi} (n_x^2 - n_y^2) \\ Y_3^3 &= \sqrt{35/32\pi} n_x (n_x^2 - 3n_y^2) \end{aligned}$$



(a) Y_2^2 (quadrupole)



(b) Y_3^3 (3-fold symmetry)

Figure 9: Real spherical harmonics. Blue = positive lobe, red = negative lobe, grey = nodal lines. Y_3^3 exhibits the 3-fold rotational symmetry of the equal-mass three-body problem.

Turing-Like Standing Waves

Three plane waves at 120° apart superpose with the 3-fold symmetry of the three-body problem:

$$v = \frac{1}{3} [\sin(fn_x) + \sin(f(\frac{1}{2}n_x + \frac{\sqrt{3}}{2}n_y)) + \sin(f(\frac{1}{2}n_x - \frac{\sqrt{3}}{2}n_y))] \quad (7)$$

Stability $\times$ Hue (Principia Encoding)

The physically motivated encoding for the Principia fractal. Hue encodes position; brightness encodes distance from the binary collision singularities:

$$L = 0.25 + 0.55 \cdot \frac{1}{2} (1 - \max_j \hat{\mathbf{n}} \cdot \hat{\mathbf{b}}_j) \quad (8)$$

where $\hat{\mathbf{b}}_j$ are the binary collision directions. Points near a collision are dark; antipodal regions are light.

Custom N -Pole VMF

N poles on a tilted great circle at elevation $\psi = \text{tilt} \cdot \pi/2$ and equatorial rotation ϕ_0 :

$$\hat{\mathbf{p}}_i = (\cos(\phi_0 + 2\pi i/N) \cos \psi, \sin(\phi_0 + 2\pi i/N) \cos \psi, \sin \psi)$$

Set $N = 3$, $\psi = 0$ to approximate three-body attractor geometry.

Basin Blend

Soft interpolation between two nearest Fibonacci cells:

$$\text{col} = t \text{ pal}[i_1] + (1 - t) \text{ pal}[i_2], \quad t = \sigma(k_s(d_1 - d_2 - 0.04))$$

Reveals near-miss geometry at basin boundaries.

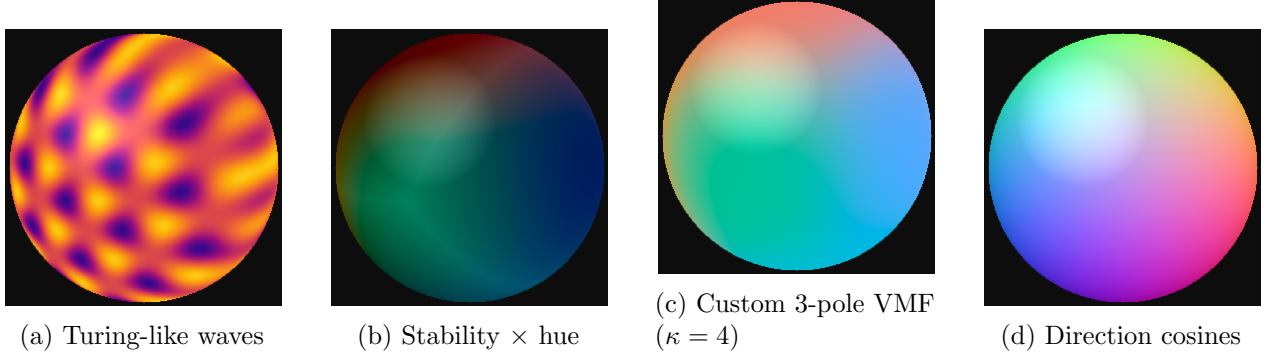


Figure 10: Special modes. The stability×hue encoding (b) is the principal Principia encoding: hue identifies basin identity while darkness marks proximity to binary collision singularities on the shape sphere.

Shape Sphere Physics Overlay

Configuration Space

For the planar equal-mass three-body problem, Montgomery’s shape sphere S^2 parametrises relative configurations modulo scale and rotation. Three classes of special configurations are overlaid:

Symbol	Configuration	Location	Count
$\text{BC}_{12}, \text{BC}_{23}, \text{BC}_{31}$	Binary collisions	Equator, 120° apart	3
$\text{E}_1, \text{E}_2, \text{E}_3$	Euler collinear configs	Equator, between BCs	3
L^+, L^-	Lagrange equilateral	North & south poles	2

Explicitly (equal-mass conventions):

$$\hat{\mathbf{b}}_1 = (1, 0, 0), \quad \hat{\mathbf{b}}_2 = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right), \quad \hat{\mathbf{b}}_3 = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0\right) \quad (9)$$

$$\hat{\mathbf{e}}_j = -\hat{\mathbf{b}}_j, \quad \hat{\mathbf{l}}^\pm = (0, 0, \pm 1) \quad (10)$$

Gaussian Blob Overlay

Each special configuration blends its colour into the pixel via a vMF blob:

$$\mathbf{c}_{\text{out}} = \mathbf{c}_{\text{base}} + \sum_j w_j (\mathbf{c}_j - \mathbf{c}_{\text{base}}), \quad w_j = s \cdot 4 \cdot \max(0, \exp(\kappa_j(\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}_j - 1) + 0.01)) \quad (11)$$

where $s \in [0, 1]$ is user-controlled strength, $\kappa_j = 11$ for BCs and 9 for Euler/Lagrange configs.

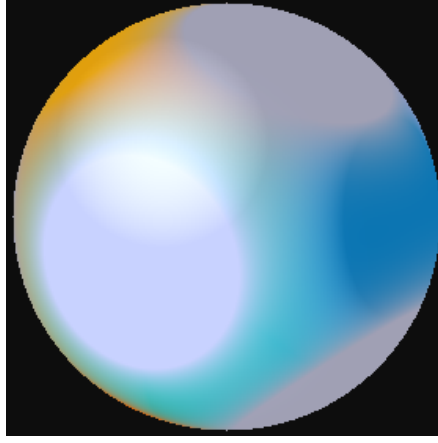


Figure 11: Shape sphere physics overlay on the Okabe–Ito VMF base map. The three binary collision points BC_{12} , BC_{23} , BC_{31} are visible as coloured blobs at the equator at 120° intervals. Lagrange equilateral configurations mark the poles. All blob colours are individually customisable.

Palette System

Fixed- N and Variable- N Palettes

Modes requiring a fixed number of colours expose individual swatches bound to native colour pickers. Variable- N modes (Fibonacci, Dot lattice, Basin blend) use a generated palette with three strategies:

1. **Golden angle.** $\theta_i = 2\pi\{i\phi_g\}$ where $\phi_g = (\sqrt{5} - 1)/2$. Adjacent indices are $\approx 137.5^\circ$ apart in hue, ensuring high contrast between neighbouring cells and CB-friendliness.
2. **Okabe–Ito cycle.** OI8 repeated. Safe for all CVD types.
3. **Gradient.** $\mathbf{c}_i = (1 - i/(N - 1))\mathbf{c}_A + (i/(N - 1))\mathbf{c}_B$.

Cubehelix Analytical Generation

For $t \in [0, 1]$, with $s = 0.5$, $\lambda = 1.5$, $h = 1$:

$$\phi = 2\pi(s/3 - \lambda t), \quad a = h \cdot t(1 - t)/2 \quad (12)$$

$$R = t + a(-0.14861 \cos \phi + 1.78277 \sin \phi) \quad (13)$$

$$G = t + a(-0.29227 \cos \phi - 0.90649 \sin \phi) \quad (14)$$

$$B = t + a(1.97294 \cos \phi) \quad (15)$$

Implementation Notes

Seam-Free Guarantee

A pixel function is seam-free on S^2 iff it is continuous. The following modes guarantee this:

- **All vMF modes:** $\exp(\kappa \hat{\mathbf{n}} \cdot \hat{\mathbf{p}})$ is smooth everywhere; the weighted mean preserves smoothness.
- **Seamless LUT sphere:** same argument applied in RGB.
- **Latitude stripes:** $\cos(f \arccos n_z)$ is smooth on S^2 .
- **Longitude stripes:** $\sin(f \operatorname{atan2}(n_y, n_x))$ is continuous at the antimeridian for integer f .
- **Perlin noise:** 3-D Cartesian noise has no seam.

Intentionally discontinuous (hard boundaries): Octant, Voronoi 6, Hemispheres, Icosahedral, Fibonacci (hard), Checkerboard, Truchet.

CB Safety Principles

1. A mode is **safe** if the deuteranopia and protanopia simulations preserve all discriminable pairs.
2. The red–green axis is unavailable under deuteranopia/protanopia ($\sim 8\%$ of males combined).
3. The blue–yellow axis is preserved across all common CVD types.
4. Sequential LUTs with monotone lightness (Viridis, Cividis, Magma, Cubehelix) are partially greyscale-readable.
5. Cividis is optimised so deuteranopes and trichromats perceive identical orderings.

Performance Notes

Sphere canvas: 280×280 ($\approx 58,000$ disc pixels). Equirectangular: 560×280 (156,800 pixels). Fibonacci/dot lattice at $N = 96$ requires $\approx 5.6 \times 10^6$ dot products per sphere frame; disable auto-rotate for high- N modes.

Quick Reference

Mode	Encoding	Perceptually uniform	CB safety
VMF OKLAB	Full 2-D	✓	Deutan issues
VMF Okabe–Ito	Full 2-D	✓	Safe (all types)
Viridis, Cividis	Full (LUT)	✓	Safe / Optimised
Plasma, Magma, Inferno	Full (LUT)	✓	Partial
Twilight	Full (cyclic)	✓	Partial
Cubehelix	Full (LUT)	Monotone L	Tolerant
Direction cosines	Full	×	Better greyscale
Voronoi/Octant	Full (discrete)	×	Depends on palette
Fibonacci	Full (discrete)	×	Safe (golden/OI)
Stability×hue	Full + physics	✓	Safe
Harmonics Y_ℓ^m	Partial	Semi	Safe

Parameter	Range	Effect
L	[0.35, 0.90]	OKLAB lightness; reserved for stability scalar
C	[0.05, 0.22]	OKLAB chroma; controls colour vibrancy
κ	[0.5, 12]	vMF concentration; blob spread
f	[2, 14]	Stripe/grid frequency (integer for seam-free)
N	[12, 96]	Fibonacci/dot cell count
k_s	[1, 20]	Sigmoid sharpness (soft Voronoi, basin blend)
s	[0, 1]	Physics overlay blob strength